



**STEEL AUTHORITY OF INDIA LIMITED
BHILAI STEEL PLANT
RESEARCH & CONTROL LAB (MECHANICAL TESTING LAB)**

Page No :A-1 of 2

IS : 2062



CM/L-0057534
MTL-II/QR-2

**TEST CERTIFICATE FOR HOT ROLLED MEDIUM AND HIGH TENSILE STRUCTURAL
STEEL**

Sold-to-Party:
BM BSO NAGPUR
345, SREE MOHINI COMPLEX, KINGS WAY
NAGPUR 440001
Ship-to-Party:
WHM NAGPUR, Nagpur.

T.C.No : RCL/MTL/PLM/80936614
T.C. Date : 28.04.2026
SALES ORDER NO : 1100585210
DA No : **80936614**

Wagon No / Trailer No : NCR21130750399

23

Process of Manufacture of Steel : Basic Oxygen Converter> Continuous Cast> Fully Killed Steel

Specification : IS 2062 E 350 BR : 2011 RA 2021

We certify that the material described below fully conforms to IS 2062 : 2011 RA 2021. Chemical composition and Mechanical properties of the product, as tested in accordance with the scheme of testing and inspection contained in the BIS certification mark Number CM/L 0057534 are as indicated against each Order No.

(Please refer to IS 2062 : 2011 RA 2021 for details of specification requirements)

TEST RESULTS

SECTION (NOMINAL SIZE)			Plate No / Pc. No.	Heat No	MECHANICAL PROPERTIES										Bend Test	M Dia
THICK	WIDTH	LENGTH			YS	UTS	%E	%E2	CHARPY V NOTCH (L) Impact Values (J)							
mm	mm	mm			330 MPa Min	490 MPa	Min22(5.6 5SqrtS0)		I1	I2	I3	Iavg	°C			
20	2500	12000	3535303/1	412037	402	510	24	--	--	--	--	--	RT	OK	2.0 T	
				412037	428	566	23	--	112	108	116	112	RT	OK	2.0 T	
20	2500	12000	3535303/2	412037	--	--	--	--	--	--	--	--	--	--	--	
20	2500	12000	3518740/1	413211	431	560	24	--	--	--	--	--	RT	OK	2.0 T	
				413211	448	573	23	--	84	88	92	88	RT	OK	2.0 T	
20	2500	12000	3518740/2	413211	--	--	--	--	--	--	--	--	--	--	--	
20	2500	12000	3530741/6	414713	450	559	24	--	94	88	92	91	RT	OK	2.0 T	
				414713	442	562	23	--	--	--	--	--	--	OK	2.0 T	
20	2500	12000	3537788/1	415763	420	543	24	--	48	46	52	49	RT	OK	2.0 T	
				415763	413	536	24	--	--	--	--	--	--	OK	2.0 T	
20	2500	12000	3537790/1	415763	--	--	--	--	--	--	--	--	--	--	--	
20	2500	12000	3537790/2	415763	--	--	--	--	--	--	--	--	--	--	--	
20	2500	12000	3537791/1	415763	--	--	--	--	--	--	--	--	--	--	--	
20	2500	12000	3537791/2	415763	--	--	--	--	--	--	--	--	--	--	--	
20	2500	12000	3537792/2	415763	--	--	--	--	--	--	--	--	--	--	--	
20	2500	12000	3537793/1	415763	--	--	--	--	--	--	--	--	--	--	--	
Heat No	CHEMICAL COMPOSITION (LADLE ANALYSIS)															
	C	S	P	Mn	Si	Al	Cu	Cr	Ni	Mo	Nb	V	Ti	N ₂	B	CE
	%	%	%	%	%	%	%	%	%	%	%	%	%	ppm	%	%
412037	0.17	0.013	0.020	1.42	0.29	0.022	< 0.02	< 0.015	< 0.015	< 0.015	< 0.010	0.048	< 0.005	62	< 0.0005	0.42
413211	0.18	0.023	0.020	1.46	0.28	0.019	< 0.02	< 0.015	< 0.015	< 0.015	< 0.010	0.041	< 0.005	64	< 0.0005	0.43
414713	0.17	0.024	0.022	1.36	0.26	0.025	< 0.02	< 0.015	< 0.015	< 0.015	< 0.010	0.043	< 0.005	63	< 0.0005	0.41
415763	0.17	0.027	0.039	1.35	0.23	0.033	< 0.02	0.020	< 0.015	< 0.015	< 0.010	0.046	< 0.005	58	< 0.0005	0.40

Total Pieces Page / Sales Order : 12 / 12

Total Pieces / TC

: 13

Signature Not Verified

Digitally signed by SSAIJU
Date: 2026.04.28 11:04:37 IST
Reason: Authorized Signatory RCL, BSP

Authorised Signatory
SAIL/ CMO/ BSO

Authorized Signatory
Research & Control Laboratory, Bhilai Steel Plant



**STEEL AUTHORITY OF INDIA LIMITED
BHILAI STEEL PLANT
RESEARCH & CONTROL LAB (MECHANICAL TESTING LAB)**

Page No :A-1 of 2

IS : 2062



CM/L-0057534

MTL-II/QR-2

**TEST CERTIFICATE FOR HOT ROLLED MEDIUM AND HIGH TENSILE STRUCTURAL
STEEL**

Sold-to-Party:

BM BSO JAIPUR

S-16A LANDMARK 4TH FLOOR OPP:GUJARATI SAMAJ MAHAVIR

JAIPUR 302001

Ship-to-Party:

WHM JAIPUR,Jaipur.

T.C.No

: 'RCL/MTL/PLM/80931377

T.C. Date

: 24.03.2026

SALES ORDER NO

: 1100584216

DA No

: **80931377**

Wagon No / Trailer No

: ER26022610034

Process of Manufacture of Steel : Basic Oxygen Converter> Continuous Cast> Fully Killed Steel

Specification : IS 2062 E 350 BR : 2011 RA 2021

We certify that the material described,below fully conforms to IS 2062 : 2011 RA 2021. Chemical composition and Mechanical properties of the product, as tested in accordance with the scheme of testing and inspection contained in the BIS certification mark Number CM/L 0057534 are as indicated against each Order No.

(Please refer to IS 2062 : 2011 RA 2021 for details of specification requirements)

TEST RESULTS

SECTION (NOMINAL SIZE)			Plate No / Pc. No.	Heat No	MECHANICAL PROPERTIES										Bend Test	M Dia
THICK	WIDTH	LENGTH			YS	UTS	%E	%E2	CHARPY V NOTCH (L) Impact Values (J)							
mm	mm	mm			330 MPa Min	490 MPa Min	Min22(5.6 5SqrtS0)		I1	I2	I3	Iavg	°C			
20	2500	12000	3517590/2	413215	410	545	23	--	88	90	102	93	RT	OK	2.0 T	
				413215	422	546	24	--	--	--	--	--	--	OK	2.0 T	
20	2500	12000	3517575/2	413225	430	555	24	--	90	98	86	91	RT	OK	2.0 T	
				413225	425	546	24	--	--	--	--	--	--	OK	2.0 T	
20	2500	12000	3517582/1	413225	--	--	--	--	--	--	--	--	--	--	--	
20	2500	12000	3517598/1	413228	422	542	24	--	94	88	90	91	RT	OK	2.0 T	
				413228	423	544	23	--	--	--	--	--	--	OK	2.0 T	
20	2500	12000	3517601/1	413228	--	--	--	--	--	--	--	--	--	--	--	
20	2500	12000	3518653/1	413331	447	566	25	--	116	110	118	115	RT	OK	2.0 T	
				413331	432	568	23	--	--	--	--	--	--	OK	2.0 T	
20	2500	12000	3518655/2	413331	--	--	--	--	--	--	--	--	--	--	--	
20	2500	12000	3518659/1	413331	--	--	--	--	--	--	--	--	--	--	--	
Heat No	CHEMICAL COMPOSITION (LADLE ANALYSIS)															
	C	S	P	Mn	Si	Al	Cu	Cr	Ni	Mo	Nb	V	Ti	N ₂	B	CE
	%	%	%	%	%	%	%	%	%	%	%	%	%	ppm	%	%
413215	0.17	0.020	0.023	1.40	0.28	0.030	< 0.02	< 0.015	< 0.015	< 0.015	< 0.010	0.046	< 0.005	65	< 0.0005	0.41
413225	0.17	0.018	0.020	1.37	0.28	0.030	< 0.02	< 0.015	< 0.015	< 0.015	< 0.010	0.051	< 0.005	64	< 0.0005	0.41
413228	0.17	0.020	0.019	1.41	0.25	0.023	< 0.02	< 0.015	< 0.015	< 0.015	< 0.010	0.044	< 0.005	69	< 0.0005	0.41
413331	0.17	0.024	0.018	1.40	0.28	0.033	< 0.02	< 0.015	< 0.015	< 0.015	< 0.010	0.051	< 0.005	66	< 0.0005	0.41

Total Pieces Page / Sales Order : 8 / 8

Total Pieces / TC

: 12

The Material supplied conforms to the standard rolling and weight tolerances..

Validity unknown

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Date: 2026.03.24 18:56:19 IST
Reason: Authorized Signatory RCL,
BSP

Authorised Signatory
SAIL/ CMO/ BSO

Authorized Signatory
Research & Control Laboratory, Bhilai Steel Plant



STEEL AUTHORITY OF INDIA LIMITED
BHILAI STEEL PLANT
RESEARCH & CONTROL LAB (MECHANICAL TESTING LAB)

Page No :A-2 of 2

IS : 2062



CM/L-0057534

MTL-II/QR-2

TEST CERTIFICATE FOR HOT ROLLED MEDIUM AND HIGH TENSILE STRUCTURAL
STEEL

Sold-to-Party:

BM BSO JAIPUR

S-16A LANDMARK 4TH FLOOR OPP.GUJARATI SAMAJ MAHAVIR

JAIPUR 302001

Ship-to-Party:

WHM JAIPUR,Jaipur.

T.C.No : RCL/MTL/PLM/80931377

T.C. Date : 24.03.2026

SALES ORDER NO : 1100584216

DA No : 80931377

Wagon No / Trailer No : ER26022610034

Process of Manufacture of Steel : Basic Oxygen Converter> Continuous Cast> Fully Killed Steel

Specification : IS 2062 E 350 BR : 2011 RA 2021

CE calculated by formula: $[C + Mn/6 + (Cr + Mo + V)/5 + (Ni + Cu)/15]$.

RT- Room temperature as defined in the specification,

Size of impact test specimen:10x10x55 mm(above 10mm thickness); 7.5x10x55 mm(upto 10mm thickness),

Note:** M = Mandrel Dia, T = Plate Thickness, Length is the rolling direction of the plate,

TC as per BSEN 10204 Type 3.1,

Validity unknown

Digitally signed by SAILJU
Date: 2026.03.24 18:56:19 IST
Reason: Authorized Signatory RCL,
BSP

Authorised Signatory
SAIL/ CMO/ BSO

Authorized Signatory
Research & Control Laboratory, Bhilai Steel Plant



STEEL AUTHORITY OF INDIA LIMITED
BHILAI STEEL PLANT
RESEARCH & CONTROL LAB (MECHANICAL TESTING LAB)

Page No :B-1 of 1

IS : 2062



CM/L-0057534

MTL-II/QR-2

TEST CERTIFICATE FOR HOT ROLLED MEDIUM AND HIGH TENSILE STRUCTURAL STEEL

Sold-to-Party:

BM BSO JAIPUR

S-16A LANDMARK 4TH FLOOR OPP:GUJARATI SAMAJ MAHAVIR

JAIPUR 302001

Ship-to-Party:

WHM JAIPUR,Jaipur.

T.C.No

: RCL/MTL/PLM/80931377

T.C. Date

: 24.03.2026

SALES ORDER NO

: 1100584217

DA No

: **80931377**

Wagon No / Trailer No

: ER26022610034

Process of Manufacture of Steel : Basic Oxygen Converter> Continuous Cast> Fully Killed Steel

Specification : IS 2062 E 350 BR : 2011 RA 2021

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(Please refer to IS 2062 : 2011 RA 2021 for details of specification requirements)

TEST RESULTS

SECTION (NOMINAL SIZE)			Plate No / Pc. No.	Heat No	MECHANICAL PROPERTIES										Bend Test	M Dia	
THICK	WIDTH	LENGTH			YS	UTS	%E	%E2	CHARPY V, NOTCH (L) Impact Values (J)								
mm	mm	mm			330 MPa Min	490 MPa Min	Min22(5.6 5Sqrt(S0))		I1	I2	I3	Iavg	°C				
25	2500	12000	3517780/1	413670	447	575	23	--	98	90	88	92	RT	OK	2.0 T		
				413670	434	560	24	--	--	--	--	--	--	OK	2.0 T		
25	2500	12000	3517790/1	413670	--	--	--	--	--	--	--	--	--	--	--		
25	2500	12000	3517876/1	413845	408	529	24	--	92	96	88	92	RT	OK	2.0 T		
				413845	422	549	23	--	--	--	--	--	--	OK	2.0 T		
25	2500	12000	3517888/1	413848	416	534	23	--	100	104	102	102	RT	OK	2.0 T		
				413848	400	527	24	--	--	--	--	--	--	OK	2.0 T		
Heat No	CHEMICAL COMPOSITION (LADLE ANALYSIS)																
	C	S	P	Mn	Si	Al	Cu	Cr	Ni	Mo	Nb	V	Ti	N ₂	B	CE	
	%	%	%	%	%	%	%	%	%	%	%	%	%	ppm	%	%	
413670	0.17	0.015	0.024	1.48	0.30	0.020	< 0.02	< 0.015	< 0.015	< 0.015	< 0.010	0.044	< 0.005	68	< 0.0005	0.43	
413845	0.15	0.024	0.022	1.38	0.26	0.028	< 0.02	< 0.015	< 0.015	< 0.015	0.034	< 0.010	< 0.005	70	< 0.0005	0.38	
413848	0.17	0.030	0.025	1.35	0.22	0.033	< 0.02	< 0.015	< 0.015	< 0.015	0.030	< 0.010	< 0.005	67	< 0.0005	0.40	

Total Pieces Page / Sales Order : 4 / 4

Total Pieces / TC

: 12

The Material supplied conforms to the standard rolling and weight tolerances.,

CE calculated by formula: $[C + Mn/6 + (Cr + Mo + V)/5 + (Ni + Cu)/15]$,

RT- Room temperature as defined in the specification,

Size of impact test specimen: 10x10x55 mm(above 10mm thickness) ; 7.5x10x55 mm(upto 10mm thickness),

Note:** M = Mandrel Dia, T = Plate Thickness, Length is the rolling direction of the plate,

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